

JavaScript

Make pages interactive · Equigrow short notes

Basics

```
let count = 0;           // can change
const name = "Asha";     // cannot be reassigned
const add = (a, b) => a + b;
const nums = [1, 2, 3];
const doubled = nums.map(n => n * 2);    // [2, 4, 6]
const even = nums.filter(n => n % 2 === 0);
const user = { name: "Asha", branch: "CSE" };
console.log(`Hi ${user.name}`);
```

- Use === (strict equals), not ==. Prefer const, then let; avoid var.

The DOM & events

```
const btn = document.querySelector("#save");
btn.addEventListener("click", () => {
  document.querySelector("#msg").textContent = "Saved!";
});
```

Fetching data (async/await)

```
async function loadUsers() {
  try {
    const res = await fetch("https://api.example.com/users");
    const data = await res.json();
    console.log(data);
  } catch (err) {
    console.error("Failed:", err);
  }
}
```

Key ideas

- Arrays: map, filter, reduce, find, forEach. Objects: key-value pairs; destructuring `const {name} = user`.
- JavaScript runs one thing at a time; async code (promises, await) waits without freezing the page.
- Use browser DevTools (F12): Console for errors, Elements for HTML/CSS, Network for requests.